

Langsi Ambe Revelation

Machine Learning in football prediction of the Belgian Jupiler pro league

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(phd)



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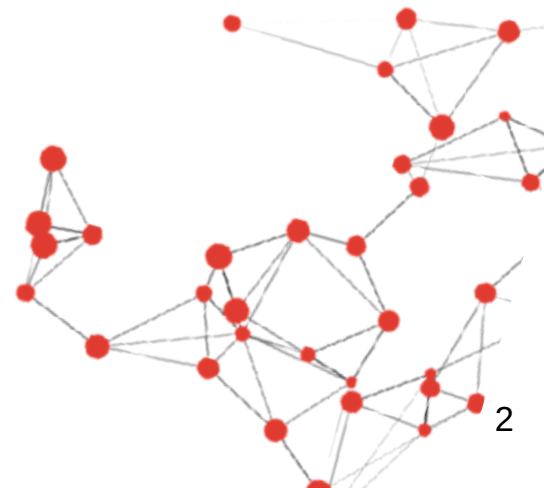
RESEARCH CONTEXT

Field of research: Data Science and Machine Learning in Sports Analytics.

Mission:

- Raise awareness about the risks of sports betting and how bookmakers manipulate odds, using AI and machine learning to educate the public.
- Enhance the understanding of the underlying factors influencing match results, including team performance, player statistics

Problem under study: The challenge of predicting football match outcomes in the Belgian Jupiler Pro League using machine learning.





RELEVANCE, STAKEHOLDERS & ETHICS

Societal relevance:

- Increase knowledge of the dangers of sports betting and how bookies tamper with odds.
- Inform the public on how betting systems are set up to favor bookmakers at the expense of bettors.

Ultimately: Assists people in making informed decisions and steer them away from the financial traps of sports betting.

Stakeholders:

- **Football fans /Bettors:** Help fans understand the true nature of sports betting, recognize odds manipulation, and reduce exposure to financial loss.
- **Coaches and Sports Analysts:** By identifying key predictor variables, coaches can use this information to enhance match performance.
- **Researchers/Academia:** Contribute to the growing field of sports analytics and ethical practices in data science, focussing on social impact and education.

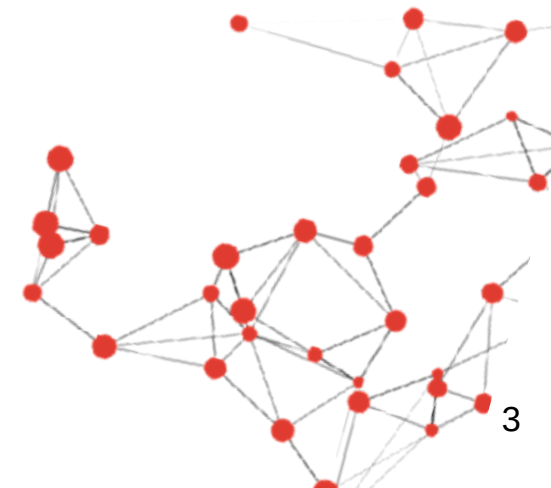
Ethical Standards:

- **Ethical approval:**

No formal approval is needed as the project uses publicly available data, but privacy and security are prioritized.

- **Integrity and Transparency**

- ❖ Ensure transparency in data analysis.
- ❖ Accurate and complete reporting of findings.





RESEARCH QUESTIONS

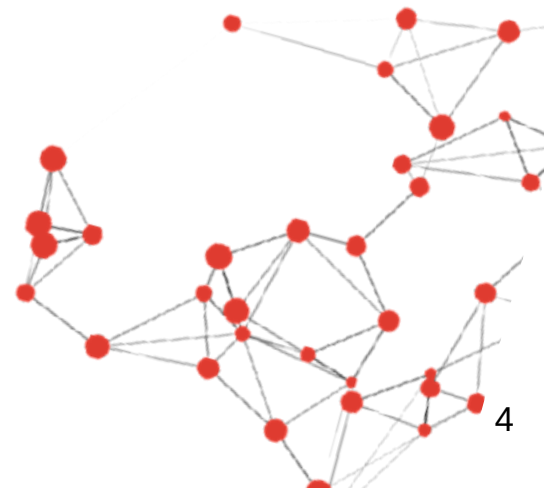
- **Priority:**

How accurately can machine learning models predict the outcomes of football matches?

What are the key features influencing match outcomes?

- **Non-priority:** Can machine learning models detect potential betting odds manipulations?

- **Additional Focus:** How can AI and ML be used to sensitize the public about sports betting and bookmakers' manipulation?





DATA

Football-data.co.uk

- Historical match statistics for teams: HC,AC,HS,AS,FTR,HG,AG etc
- Betting odds from different book makers

Data from 2019/2020 to 2023/2024 (5 seasons) = 1462 matches retrieved

Sofifa.com

- Overall team rating
- Team attacks rating
- Team defence rating
- Team midfield rating
- Buildup style

Fbref.com

- Team match possession
- Team match formation
- Coach
- Referee

Still Retrieving using Microsoft power query





METHODOLOGICAL APPROACH

EDA

- ☐ missing data handling
- ☐ Descriptive statistics
- ☐ Finding patterns and correlation

Feature Engineering

- ☐ Creating rolling averages of team statistics
- ☐ **Head-to-head stats**
- ☐ **Team stats**

Unsupervised Learning

- ☐ K-means clustering
- ☐ Ensemble Learning

Supervised Learning

- ☐ Logistic regression
- ☐ SVM
- ☐ Random Forest
- ☐ Xgboost
- ☐ NN
- ☐ KNN
- ☐ Naïve bayes

Model Evaluation

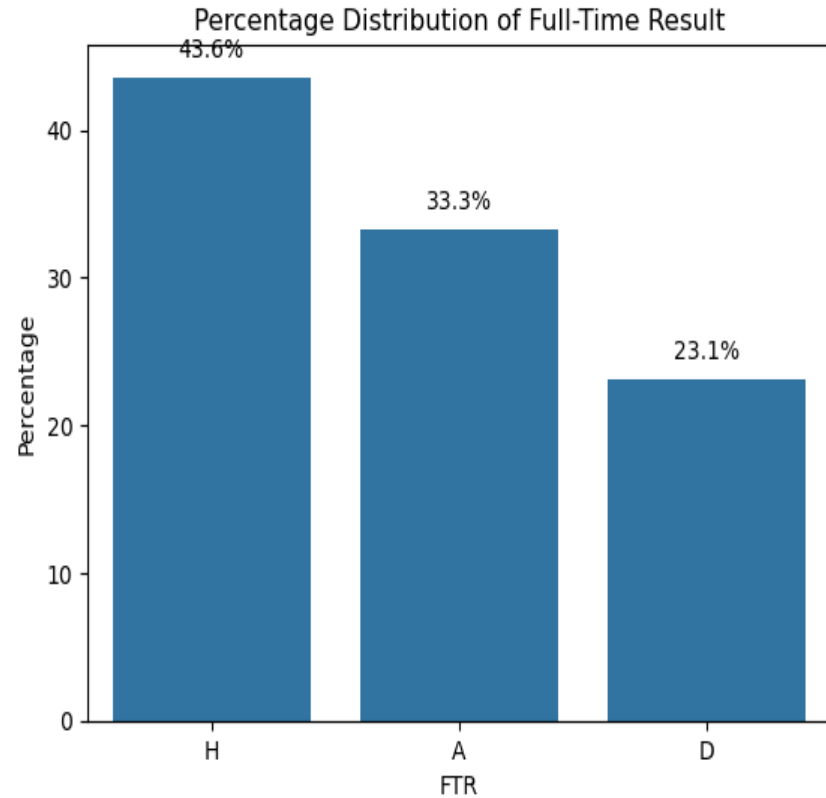
- ☐ Accuracy
- ☐ Recall
- ☐ Precision
- ☐ F1-score

Insides

- ☐ Look at best-performing Models and the important features attributed to them
- ☐ Create odds from the predictions and compare to the allocated odds.

PRELIMINARY RESULTS

Total of 22 teams in this period:

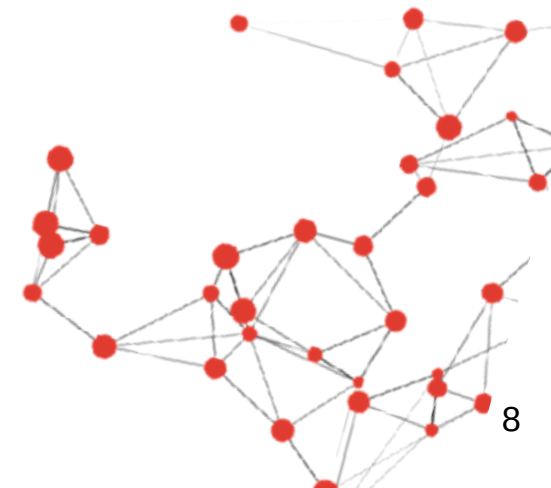
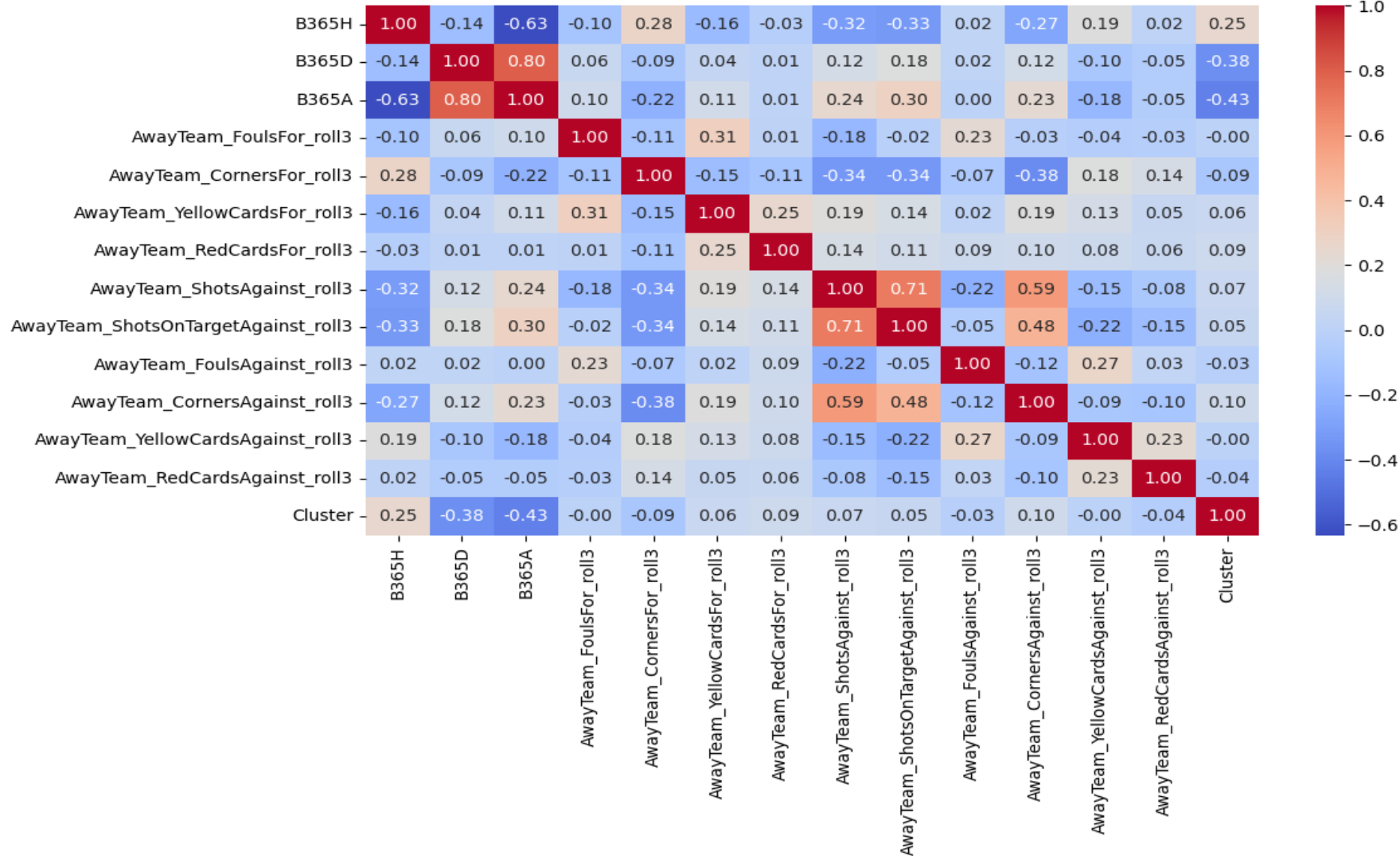


Proportion of Missing Values per Variable

Variable	%Missing
HS	0.34
AS	0.34
HST	0.34
AST	0.34
HF	0.34
AF	0.34
HC	0.34
AC	0.34
HY	0.27
AY	0.27
HR	0.27
AR	0.27
B365H	0.07
B365D	0.07
B365A	0.07

PRELIMINARY RESULTS

Correlation Heatmap of Selected Features

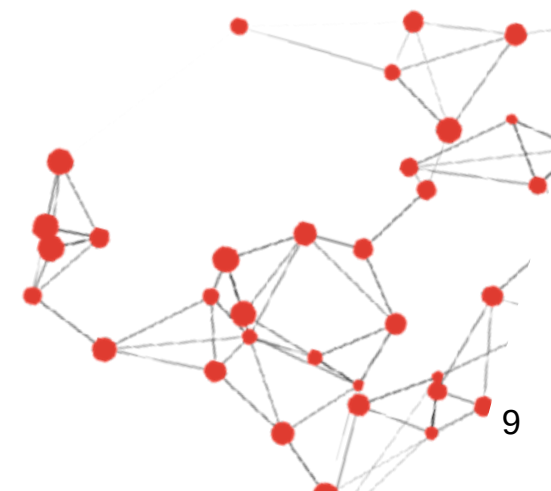




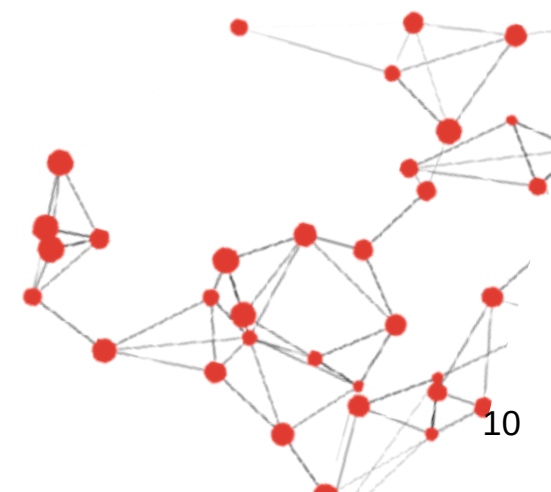
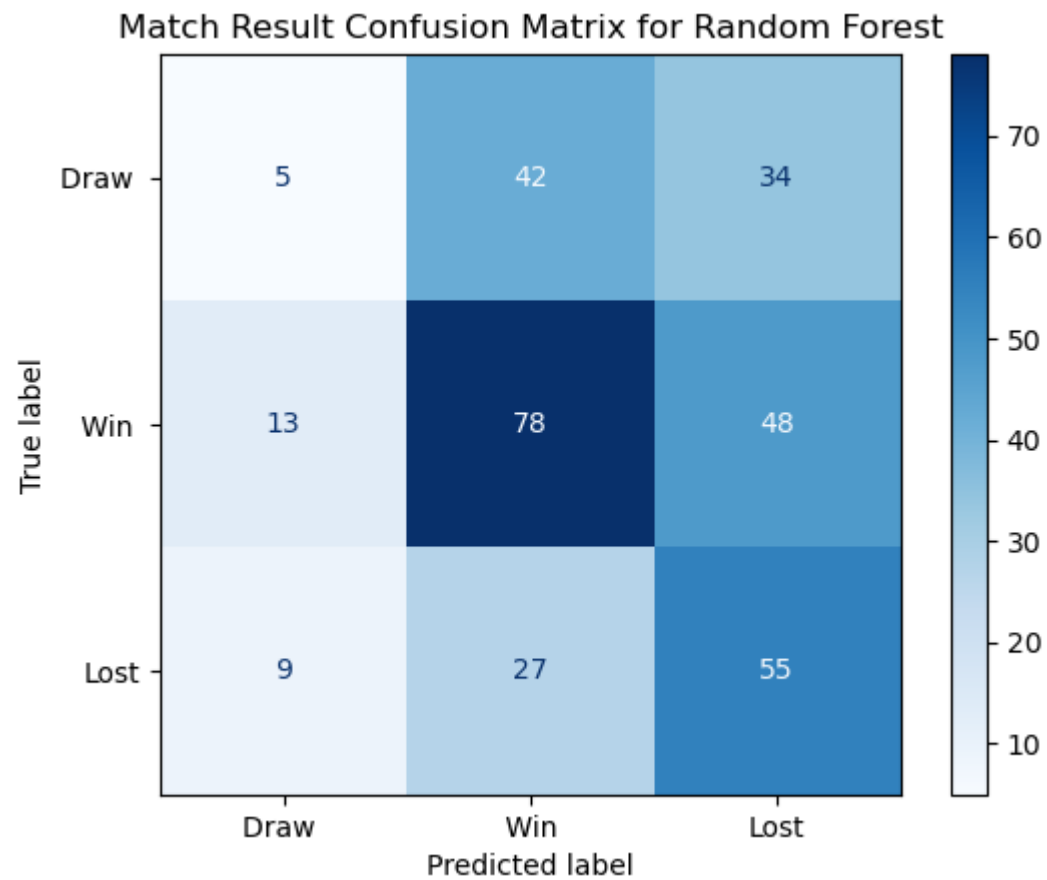
PRELIMINARY RESULTS

Model Comparison

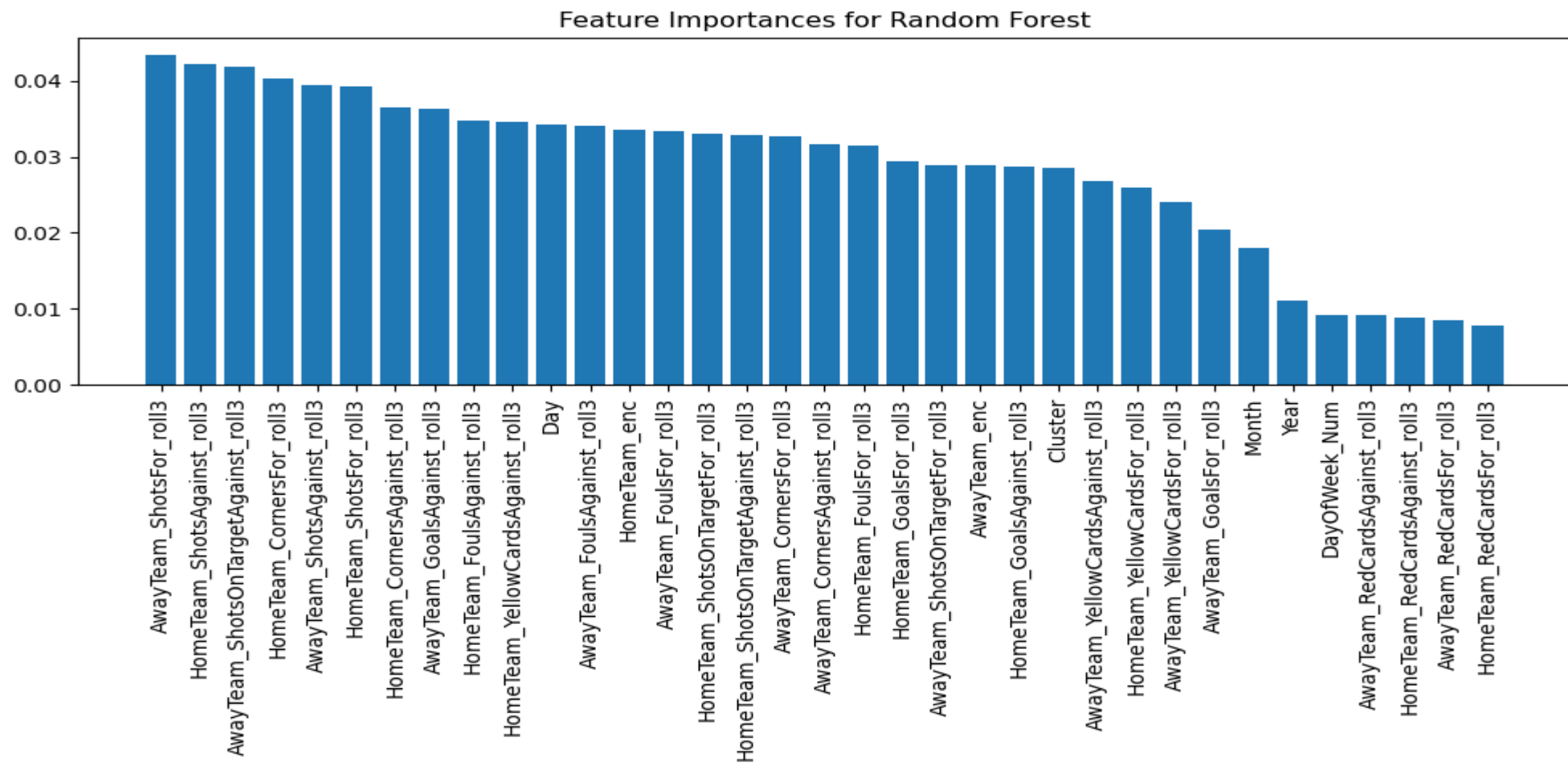
Model	Accuracy	Precision	Recall	F1 Score
Naive Bayes	0.473	0.457	0.473	0.441
K-Nearest Neighbors	0.473	0.416	0.473	0.433
Random Forest	0.498	0.440	0.498	0.432
Neural Network	0.466	0.432	0.466	0.423
Support Vector Machine	0.492	0.368	0.492	0.421
Logistic Regression	0.479	0.423	0.479	0.415
XGBoost	0.469	0.432	0.469	0.413
Decision Tree	0.357	0.368	0.357	0.360



PRELIMINARY RESULTS



PRELIMINARY RESULTS





RISK ANTICIPATION

Anticipation of problems or difficulties in the thesis

Data Availability and Quality: Inconsistent historical data and missing key features.

External Factors: Accounting for unpredictable variables (e.g., injuries, referee decisions, red cards during the game).



TIME PLAN

A few weeks of thesis work have passed already. What have you already done?

- Introduction, Literature review, Methodology(in progress).

Time	Task	Objective	% completion
Oct 06 – oct 19	Setup and data collection	Gather and clean data, setup project environments	90%
Oct 20 - Nov 02	Exploration and feature Engineering	EDA, identify trends and patterns,start feature engineering	100%
Nov 04- Nov 24	Model selection, training and feature engineering, reporting	More advance feature engineering, train baseline models and advance models, start report writing	78%
Nov 30- Dec 08	Model refinement, feature engineering, odds calculations,reporting	Generate odds from best model and start comparison to identify potential manipulations	15%
Dec 14- Dec 28	Finalize results and analysis,add visualization	Ensure coherent Narrative	0%
Dec 30-Jan 06	Thesis Reveiw	thesis review and final touches: Proofreading and seeking feedback	0%
Jan 10- Jan 20	Final submission and defense prepatation	Prepare for final submission and defense	0%

How frequently do you report to your supervisors or ask feedback?

- To my assistant as necessary and on programmed meetings with Supervisor.